# Chapter 5: Present Worth and Related Economic Criteria

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**Instructions:**

- For every calculation, clearly state which interest factor(s) you are using in standard notation (e.g., $(P/A, i, n)$) before substituting numerical values.
- Show the formula you are using before computing the final numerical answer.
- Round final dollar answers to the nearest cent, unless otherwise stated.

### Exercise 1: Classifying Alternatives

A municipal utility company is evaluating how to improve the reliability of its power distribution network. Two entirely different substation transformer systems are being considered from two different vendors; the utility can only install one of the two, since there is only space and budget for a single new unit.

Separately, the utility is also considering three smaller, unrelated upgrades to its existing grid infrastructure: replacing aging circuit breakers at one substation, installing smart meters in a residential district, and adding a backup diesel generator at a critical facility. Each of these three upgrades can be approved or rejected independently of the others, and any combination of them (including none) could be implemented alongside whichever transformer decision is made.

1. Identify which set of alternatives (the transformer systems, or the three smaller upgrades) is *mutually exclusive*, and which is a set of *independent projects*. Briefly justify your answer.
2. For each of the two groups, state whether it should be treated as a revenue alternative or a cost (service) alternative, assuming the transformers and upgrades only generate cost savings and do not generate any new revenue streams.

### Exercise 2: The AI Data Center Dilemma

A tech giant is evaluating a cutting-edge liquid immersion cooling system for its new AI data center, aiming to replace traditional HVAC methods. The new system requires an

initial capital investment of \$4,800,000.

Because the cooling fluid degrades at high operating temperatures, the system requires a specialized fluid swap every 2 years costing \$150,000. Routine maintenance is expected to cost \$15,000 per month for the first year. However, due to continuous component wear and scaling operations, this monthly cost is projected to increase by an arithmetic gradient of \$1,000 each month starting in Year 2 (i.e., Month 13 costs \$16,000, Month 14 costs \$17,000, and so on).

Despite these maintenance costs, the immersion system generates immense energy efficiency, yielding energy savings of \$80,000 per month. The cooling system will be evaluated over a 5-year life (60 months) with zero salvage value at the end. The company's Minimum Attractive Rate of Return (MARR) is a nominal 12% compounded *monthly*.

1. Determine whether the company should invest in this cooling system. Justify your decision using a comprehensive Present Worth analysis.

**Exercise 3: Independent Projects Screening with Capital Rationing**

An agricultural cooperative has identified three independent efficiency projects it could pursue to modernize its irrigation and storage systems. The cooperative's MARR is 9% per year.

- **Project 1:** Initial cost \$200,000; generates annual net savings of \$25,000 indefinitely.
- **Project 2:** Initial cost \$300,000; generates a flat annual net savings of \$65,000 for exactly 10 years, with no salvage value.
- **Project 3:** Initial cost \$150,000; generates savings of \$15,000 in Year 1, growing geometrically by 6% per year for 10 years, with no salvage value.

1. Assuming the cooperative has an unlimited budget, which project(s) should be implemented?
2. Suppose the cooperative suddenly faces a strict budget constraint of \$400,000 this year. Determine the optimal combination of projects the cooperative should select. Justify your answer.

**Exercise 4: Mutually Exclusive Revenue Alternatives with Equal Lives**

A telecommunications company is deciding which of two new network monitoring platforms to adopt; only one can be selected, and the current manual monitoring process

remains an option. Both platforms have a useful life of 5 years, and the company's MARR is 10% per year.

- **Platform X:** Costs $250,000 to implement, and is expected to generate additional annual net revenue (through reduced downtime and improved customer retention) of $58,000.
- **Platform Y:** Costs $140,000 to implement, and is expected to generate additional annual net revenue of $34,000. However, to keep up with security compliances, Platform Y requires a mandatory software upgrade costing $40,000 exactly at the end of Year 3.

1. Determine which alternative(s) the company should select using a Present Worth analysis.

**Exercise 5: Comparing Evaluation Methods: LCM vs. Study Period**

A logistics company is deciding between two mutually exclusive automated sorting systems. The company's MARR is 10% per year.

- **System A:** Initial cost $150,000. Expected useful life of 2 years. Generates annual net savings of $115,000. Salvage value is $15,000.
- **System B:** Initial cost $250,000. Expected useful life of 3 years. Generates annual net savings of $120,000. Salvage value is $30,000.

1. Determine which system should be selected using the Lowest Common Multiple (LCM) approach. Show all cash flow evaluations.
2. The company's CFO argues that due to rapid technological shifts, the investment should instead be evaluated over a strict **3-year study period**. Evaluate which system should be selected under this study period approach.
3. Does the recommended decision change between the two evaluation methods? Briefly explain conceptually why the results might differ.

**Exercise 6: Comparing Alternatives with Unequal Lives (LCM Approach)**

A mining company is comparing two mutually exclusive options for a new fleet haul truck, since only one truck is needed for the route in question. The company's MARR is 9% per year.

- **Truck Model P:** Initial cost \$180,000, useful life of 4 years. Generates annual net operating savings (compared to the current leased truck) of \$52,000. No salvage value.
- **Truck Model Q:** Initial cost \$260,000, useful life of 6 years. Generates annual net operating savings of \$68,000. It has an estimated salvage value of \$20,000 at the end of each life cycle. Additionally, Model Q requires a major engine overhaul costing \$30,000 exactly at the end of Year 3 of its life cycle.

1. Recommend which truck model the company should purchase using the Lowest Common Multiple (LCM) approach. Show all cash flow evaluations over the equal-service period.

**Exercise 7: Evaluating Alternatives Over a Fixed Study Period**

A manufacturing plant is evaluating two mutually exclusive automated assembly machines. Because rapid technological advancements are expected to render current technology obsolete soon, the plant manager mandates a strict **6-year study period** for the economic evaluation. The company's MARR is 10% per year.

- **Machine Alpha (Life = 8 years):** Costs \$3,200,000. Generates annual net savings of \$650,000. If kept for its full 8-year life, its salvage value is \$200,000. However, its estimated market value exactly at the end of Year 6 is \$900,000.
- **Machine Beta (Life = 4 years):** Costs \$1,400,000. Generates annual net savings of \$400,000. Salvage value at the end of Year 4 is \$150,000. The machine will not be replaced at Year 4.

1. Based on a Present Worth analysis over the required 6-year study period, which machine(s) should the plant manager select? Show all calculations.

**Exercise 8: Future Worth Analysis: Venture Capital Exit Strategy**

A private equity firm acquires a promising tech startup for \$10,000,000 today (Year 0). To aggressively capture market share, the firm injects an additional capital of \$2,000,000 into the startup at the end of Year 1, and another \$2,000,000 at the end of Year 2.

Due to heavy marketing, the startup generates a negative net operating cash flow of -\$1,000,000 at the end of Year 1. However, operations turn highly profitable in Year 2, generating \$1,500,000 in net profit. From Year 3 to Year 6, these net profits grow geometrically by 15% each year.

The firm's MARR is a steep 18% per year. The private equity firm plans to exit and sell the startup entirely at the end of Year 6.

1. Using Future Worth (FW) analysis, calculate the exact minimum selling price (Terminal Value) the firm must secure at the end of Year 6 in order to exactly achieve its 18% MARR on the entire investment.

**Exercise 9: Capitalized Cost: Comparing Finite vs. Infinite Facilities**

A city government is evaluating designs for a new pedestrian bridge crossing a busy highway. The city's MARR is 6% per year. Two options are proposed:

**Option A: Permanent Suspension Bridge (Infinite Life)**

- Initial construction cost: $4,500,000.
- A major structural inspection and reinforcement is required every 20 years, costing $600,000 each time.
- Routine annual maintenance costs $45,000 per year for the first 10 years, stepping up to $70,000 per year from Year 11 onward, indefinitely.
- The lighting and safety-rail systems must be fully replaced every 25 years at a cost of $250,000.

**Option B: Modular Steel Bridge (Finite Life)**

- Initial construction cost: $1,800,000.
- Expected useful life: 15 years.
- Routine annual maintenance: $120,000 per year.
- Salvage value at the end of its 15-year life: $150,000.

1. Based on a Capitalized Cost (CC) comparison, which option is the most economical choice over an infinite planning horizon? Show all calculations.

**Exercise 10: Payback Period and Its Limitations**

A commercial bakery is comparing two mutually exclusive industrial ovens for a 6-year project, using a simple (0% interest) payback period analysis as an initial screening tool.

- **Oven M:** Costs $60,000. Generates net annual cash flow of $15,000 for each of the 6 years. Has no salvage value.

- **Oven N:** Costs \$60,000. Generates net annual cash flow of \$11,000 for each of the 6 years, but is expected to have a massive salvage value of \$28,000 at the end of Year 6.

1. Which oven appears preferred based strictly on a simple payback period analysis?
2. Is this the most economical choice? Quantify the total un-discounted cash received over the full 6-year life for both ovens to explain why relying solely on a payback period analysis can be misleading.

**Multiple Choice Questions**

The following multiple-choice questions test your conceptual understanding and execution speed.

1. The Capitalized Cost (CC) of an alternative with a finite life of $n$ years and an equivalent uniform annual worth $A$ is calculated as:
   (a) $A(P/A, i, n)$
   (b) $A/i$
   (c) $A(F/A, i, n)/i$
   (d) $A \times i$
2. When comparing mutually exclusive alternatives using the lowest common multiple (LCM) approach, a fundamental assumption is that:
   (a) The cash flow estimates will continuously increase by the inflation rate in subsequent life cycles.
   (b) The alternatives will not be replaced at the end of their first life cycle.
   (c) The cash flow estimates are the exact same in every subsequent life cycle.
   (d) The MARR will strictly decrease in subsequent life cycles.
3. Excel's `NPV(rate, value1, [value2], ...)` function makes a rigid assumption about cash flow timing. It assumes that the first cash flow in the array (`value1`) occurs:
   (a) At time $t = 0$ (the present).
   (b) At the end of period 1.
   (c) At the beginning of period 1.
   (d) At the end of the study period.
4. In Engineering Economy, which of the following is true regarding **independent projects**?

(a) They must be compared over an equal-service period using the LCM approach.

(b) They compete directly with one another for selection, and only one can be chosen.

(c) Each project is evaluated independently against the do-nothing alternative.

(d) They must be treated solely as cost alternatives, never revenue alternatives.

5. When using a study period approach for a revenue alternative whose useful life is *shorter* than the study period, the assumption is typically to:

(a) Assign a market value to the alternative exactly at the end of the study period.

(b) Ignore the years between the end of the alternative's life and the end of the study period.

(c) Assume the remaining years have a heavily negative present worth to penalize the shorter life.

(d) Reinvest the accumulated net cash flows at the MARR until the end of the study period.